



Autonomous Mobile Robots and Programming

Introduction and Lecture Overview

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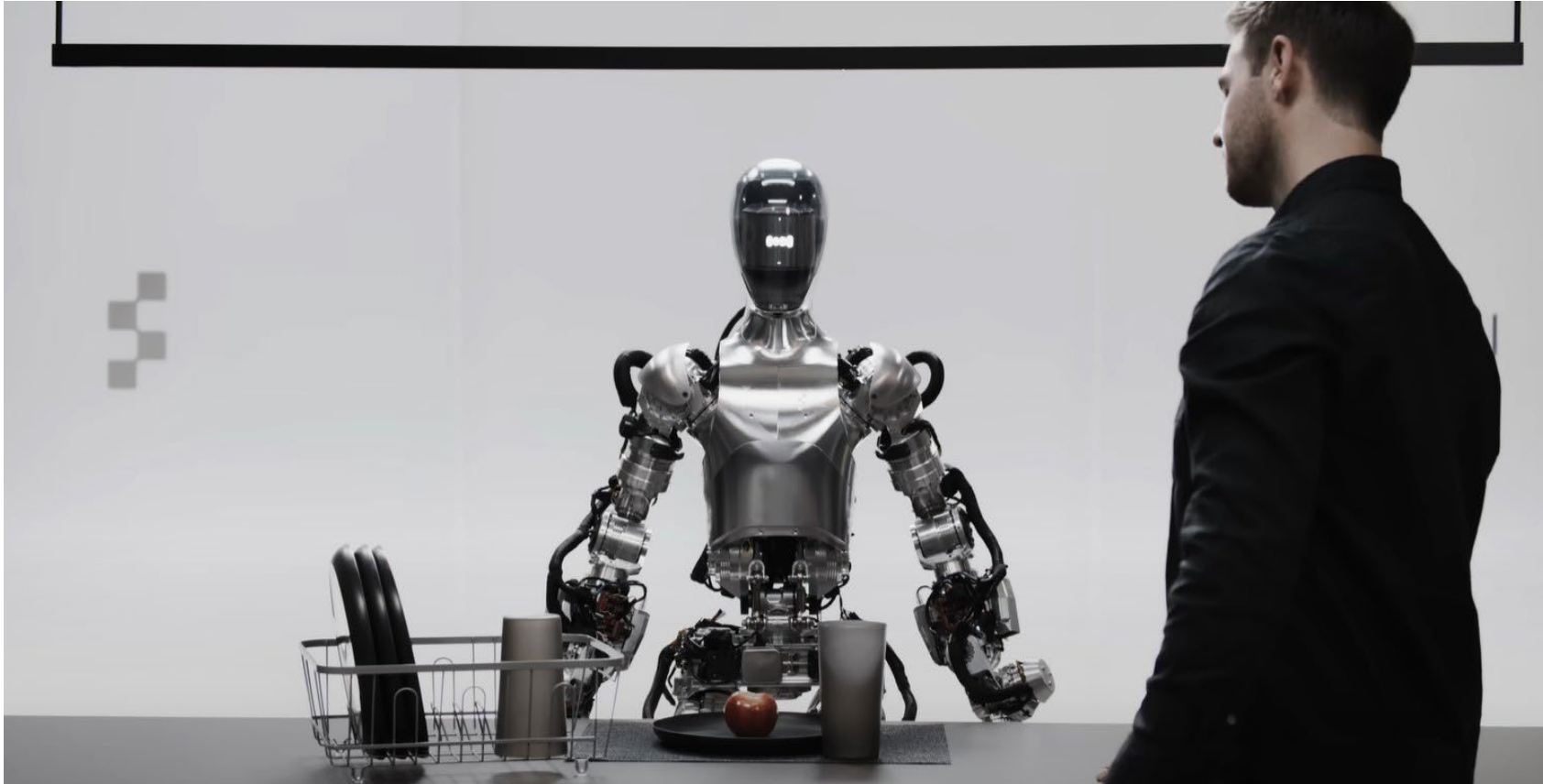
Courtesy of:

Roland Siegwart, Margarita Chli, Nick Lawrance

Assignments & Exams

- Programming assignments
- Programming projects: C++ and Python
- Final project and presentation
- Simulation platforms:
 - ARGoS: <https://www.argos-sim.info/>
 - Webots: <https://cyberbotics.com/>
 - Not decide yet since it takes a long process to get the permission of installing on CS lab computers.

Robot Demo



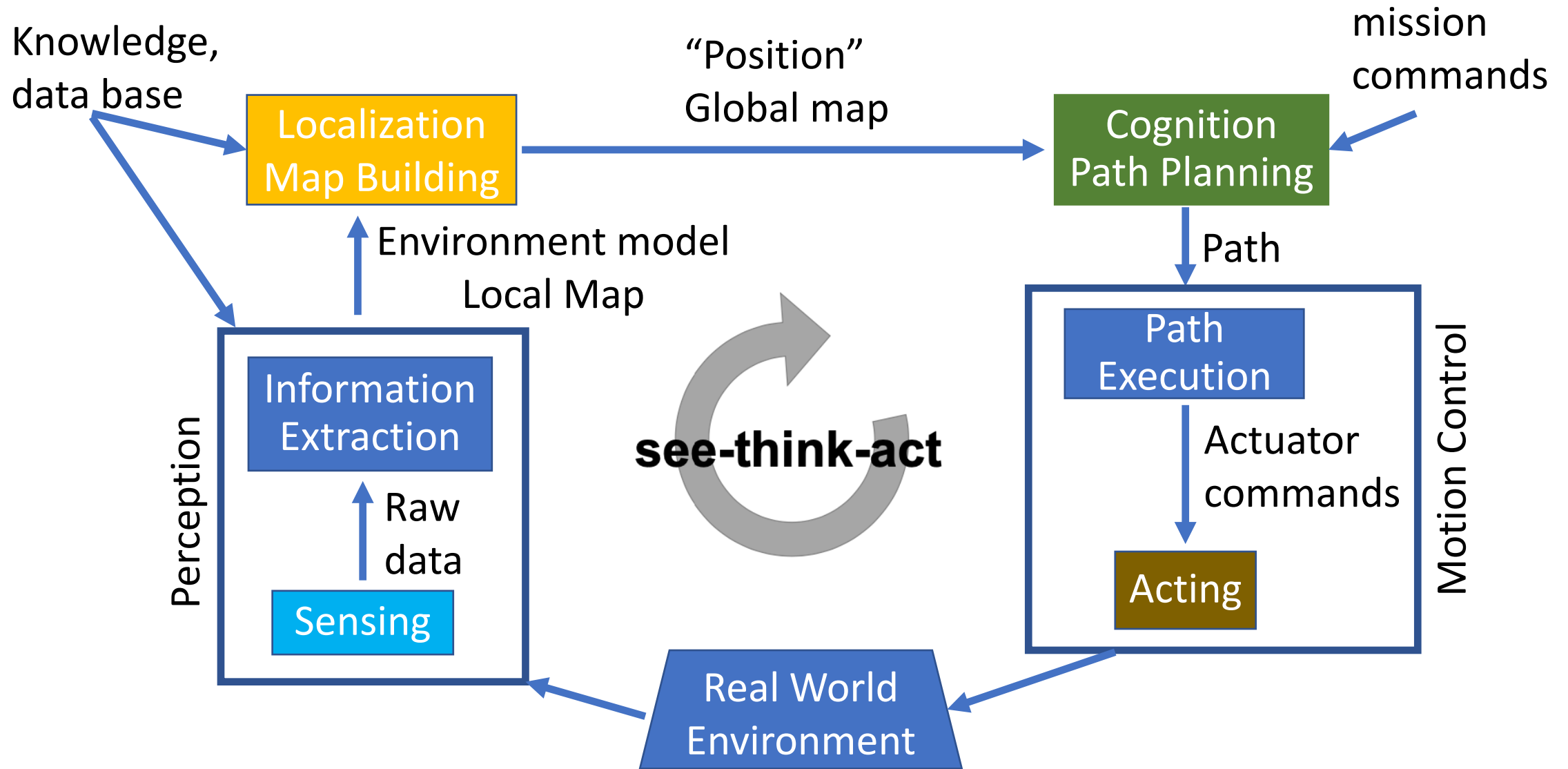
- <https://www.youtube.com/watch?v=Sq1QZB5baNw>

Autonomous Mobile Robots

- The three key questions in mobile robotics
 - Where am I?
 - Where am I going?
 - How do I get there?
- To answer these questions the robot must ...
 - Have a model of the environment (given or autonomously built)
 - Perceive and analyze the environment
 - Find its position/situation within the environment
 - Plan and execute the movement



The see-think-act cycle

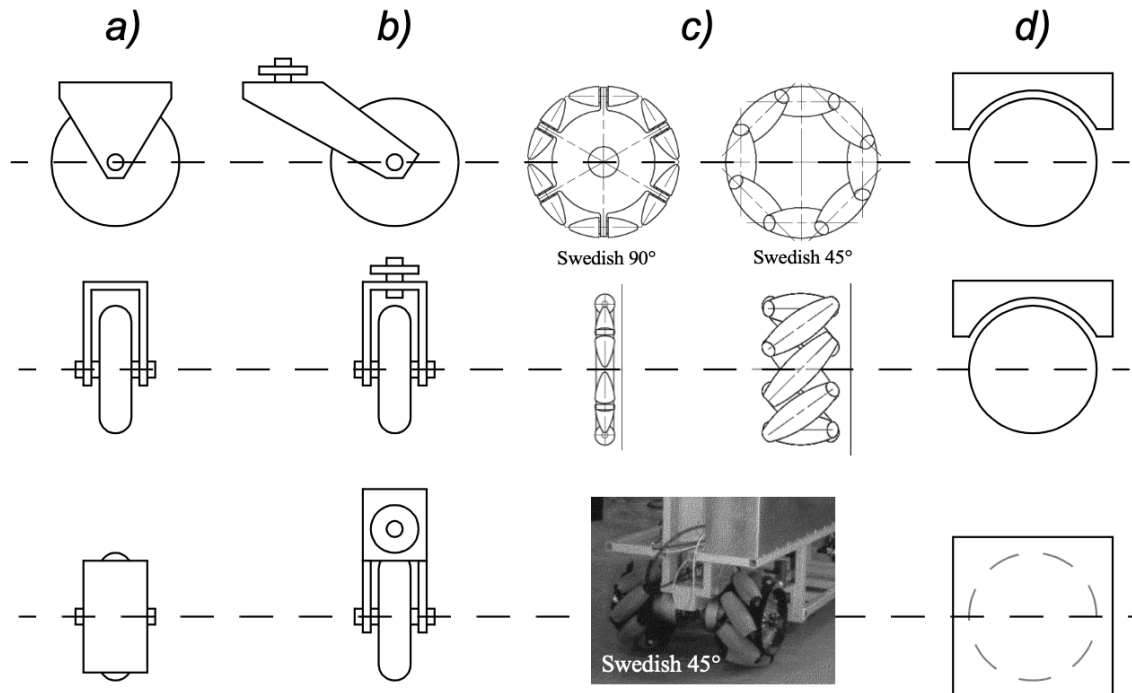


Some interesting videos in Locomotion

- Boston Dynamic:
 - <https://www.youtube.com/watch?v=uhND7Mvp3f4>
 -
- Legged and wheeled robots
 - https://www.youtube.com/watch?v=_rPvKlvyw2w
 - <https://www.youtube.com/watch?v=ukY0vyM-yfY>

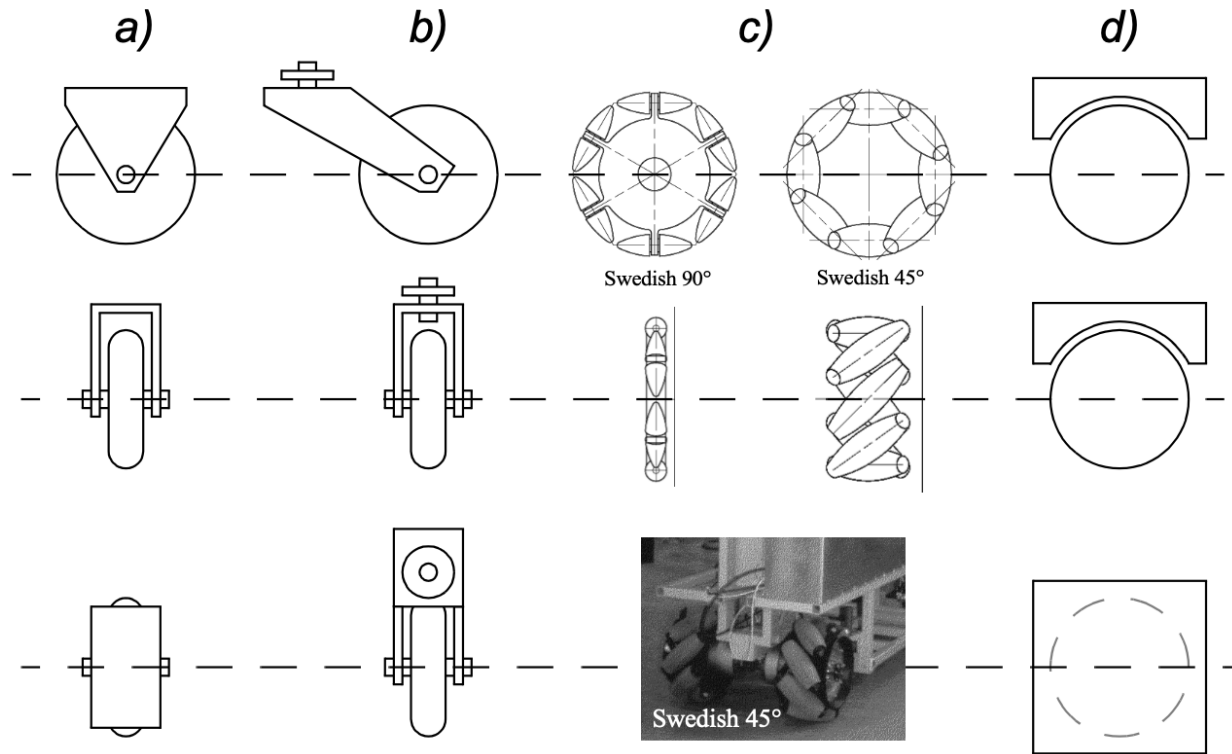
Motion Control: Kinematics & motion control

- Wheel types and its constraints
 - Rolling constraint
 - No-sliding constraint (lateral)



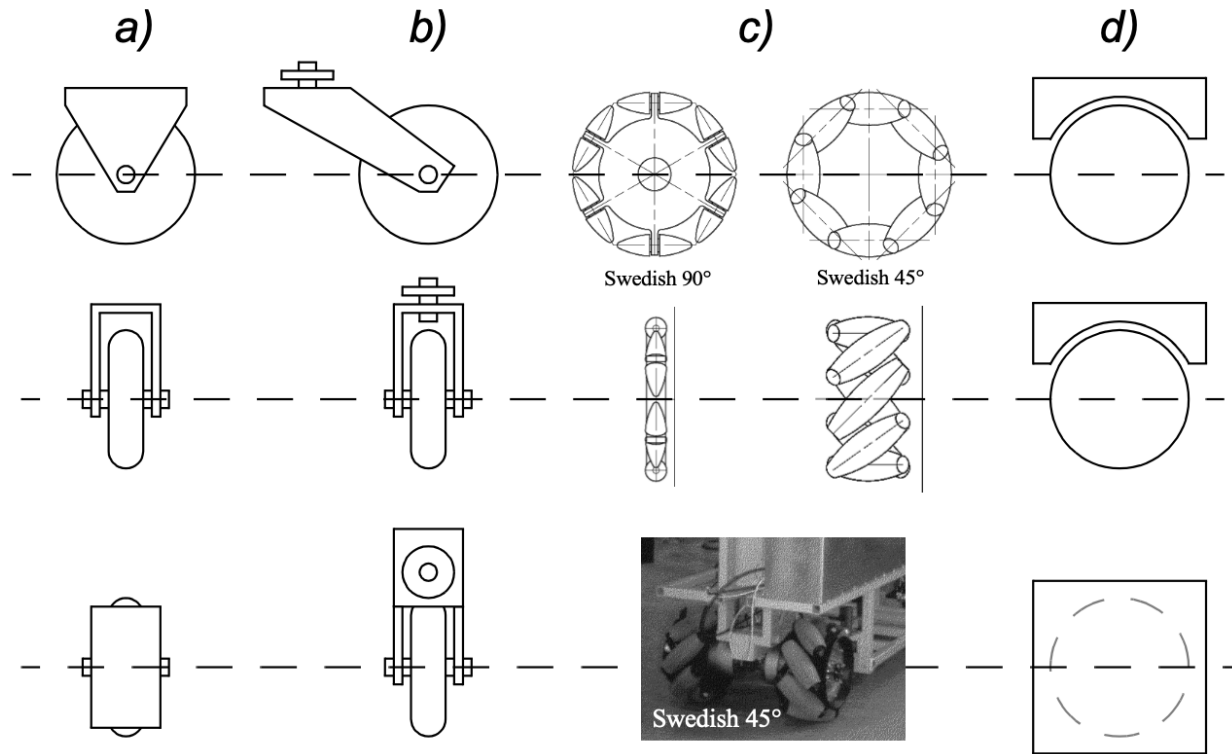
Ref. to Ch. 2

Motion Control: Kinematics & motion control



- (a) **Standard wheel:** two degrees of freedom; rotation around the (motorized) wheel axle and the contact point.
- (b) **Castor wheel:** two degrees of freedom; rotation around an offset steering joint and the contact point.

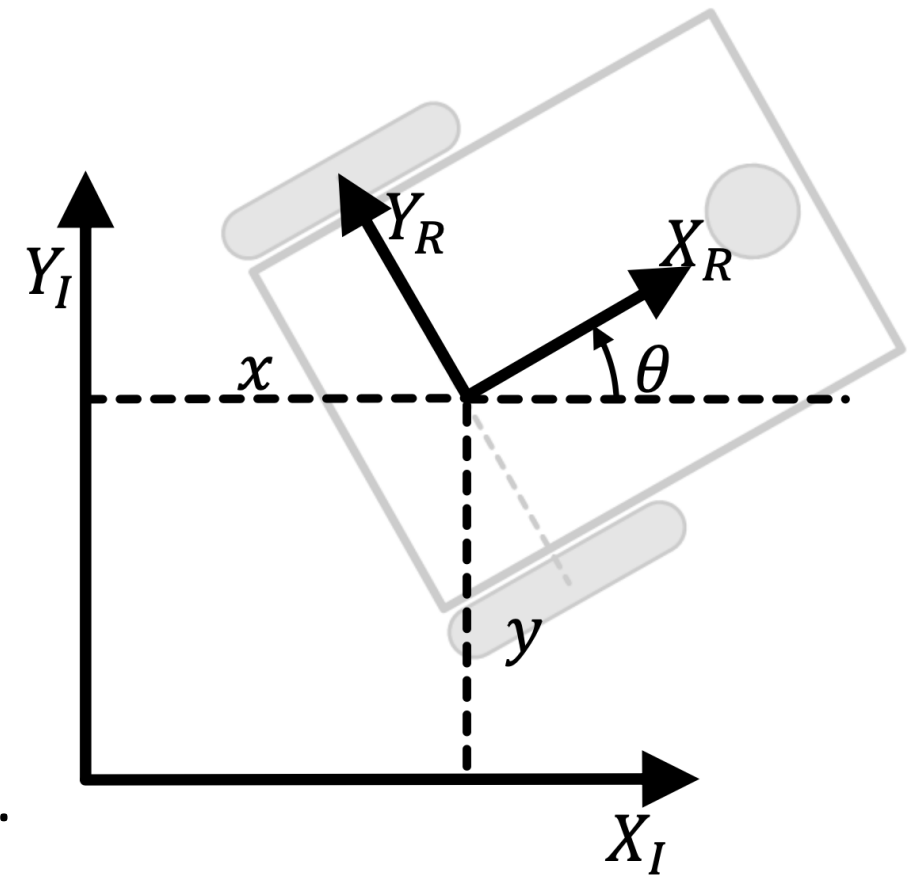
Motion Control: Kinematics & motion control



- (c) **Swedish wheel:** three degrees of freedom; rotation around the (motorized) wheel axle, around the rollers, and around the contact point.
- (d) **Ball or spherical wheel:** realization technically difficult.

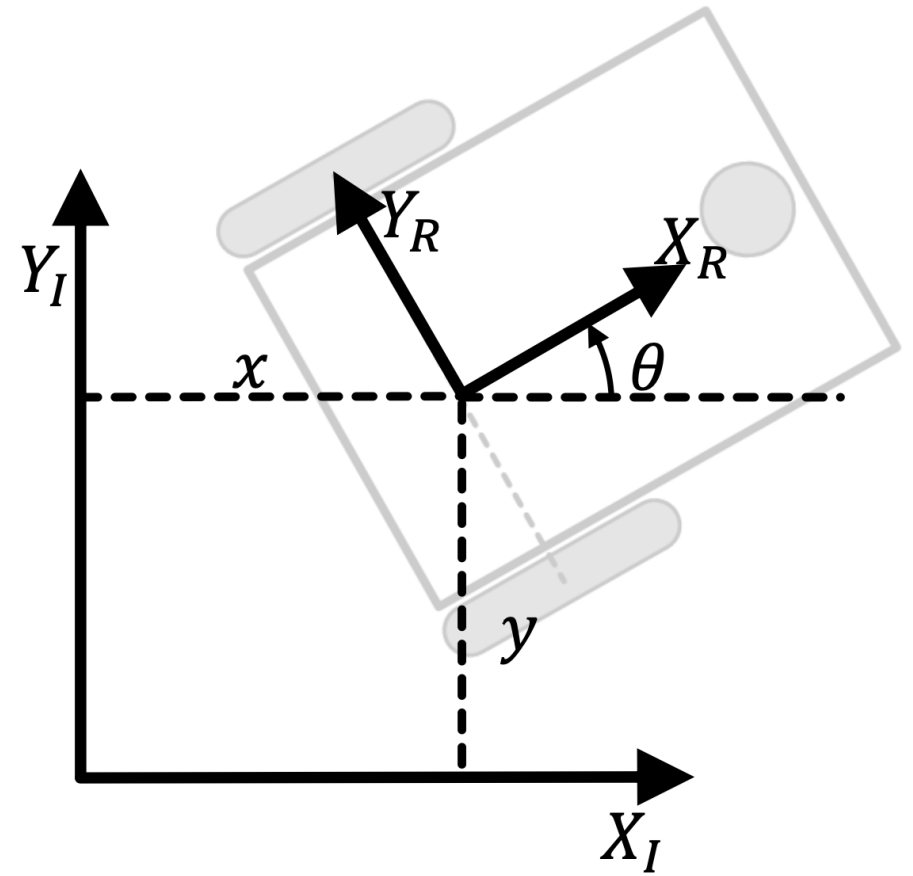
Motion Control: Kinematics & motion control

- We model the robot as a rigid body on wheels, operating on a horizontal plane.
- The total dimensionality is three, two for position in the plane and one for orientation along the vertical axis, which is orthogonal to the plane.
- Of course, there are additional DOF and flexibility due to the wheel axles, wheel steering joints, and wheel castor joints. However, by robot chassis we refer only to the rigid body of the robot, we ignore them.



Motion Control: Kinematics & motion control

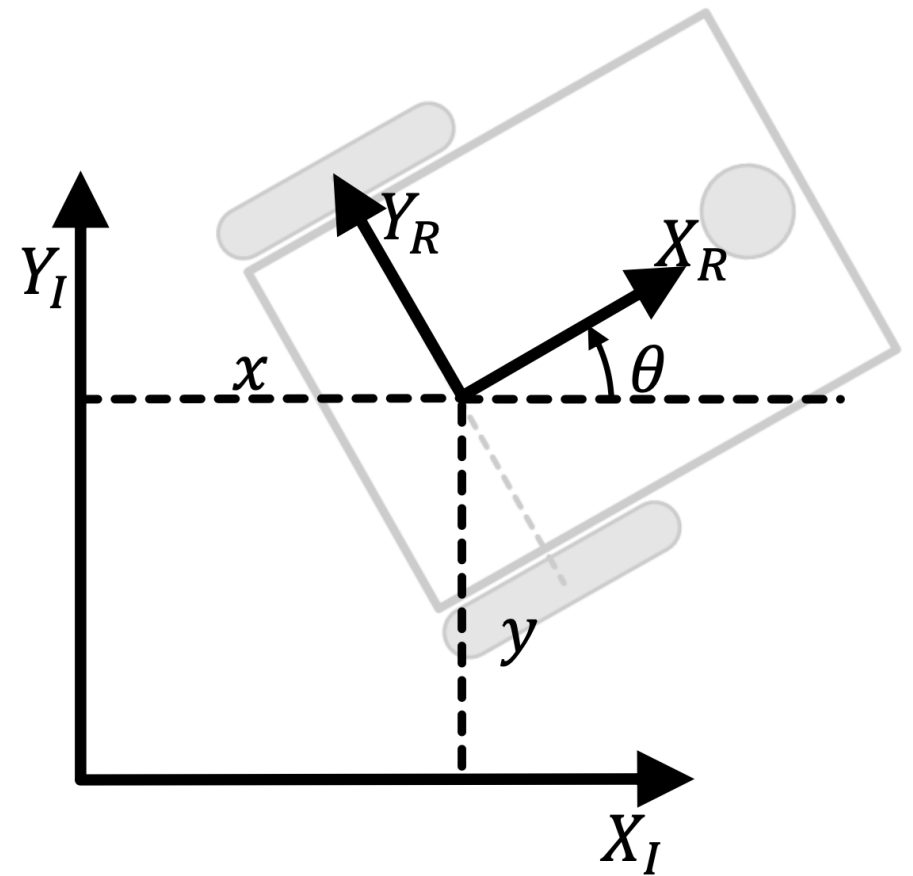
- $\xi_I = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$ is the pose of the robot in the global frame with axes X_I and Y_I .
- X_R and Y_R are two axes relative to the robot chassis and is thus the robot's local reference frame.



Motion Control: Kinematics & motion control

- It will be necessary to map motion along the axes of **the global reference frame** to motion along the axes of the robot's **local reference frame**, from $\{X_I, Y_I\}$ to $\{X_R, Y_R\}$.
- The orthogonal rotation matrix:

$$R(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

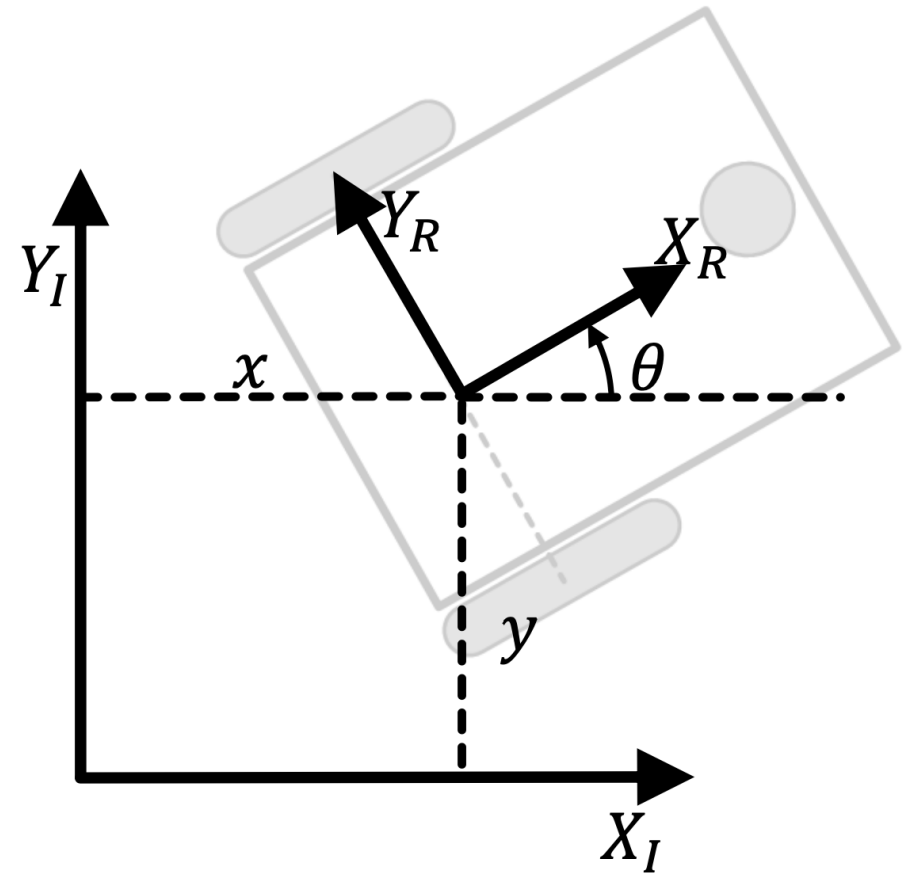


Motion Control: Kinematics & motion control

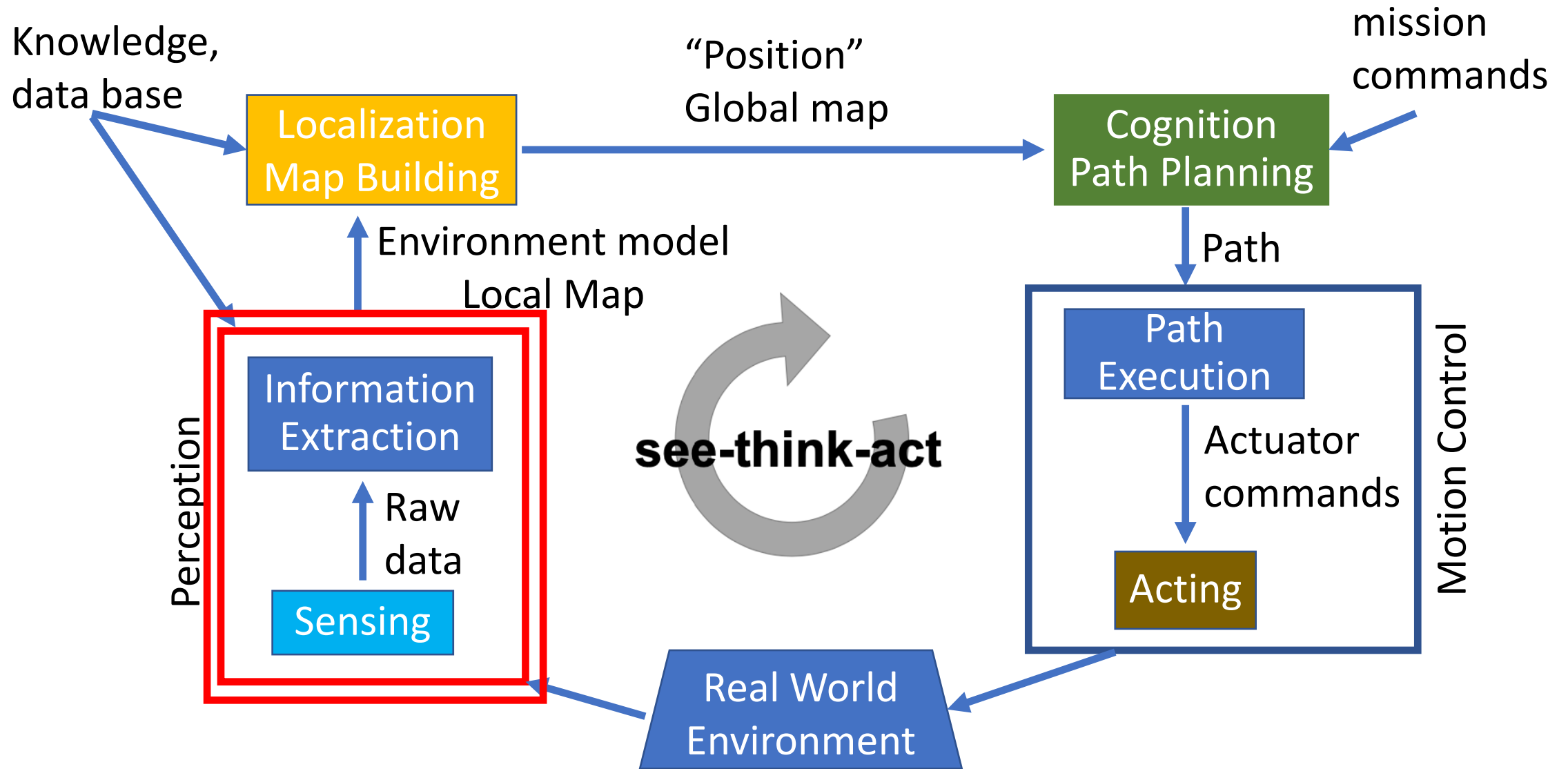
$$\xi_R = R(\theta)\xi_I$$

$$= \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}$$

$$= \begin{bmatrix} x\cos\theta + y\sin\theta \\ -x\sin\theta + y\cos\theta \\ \theta \end{bmatrix}$$



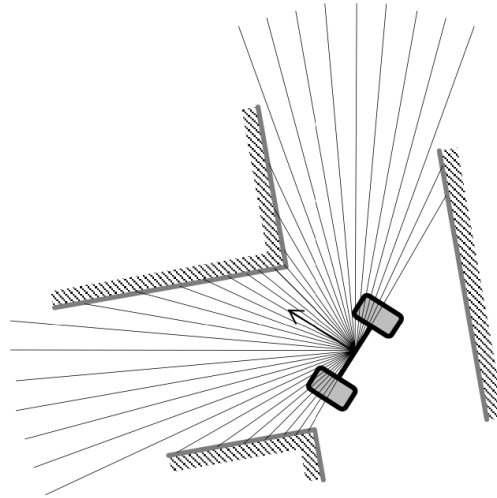
The see-think-act cycle



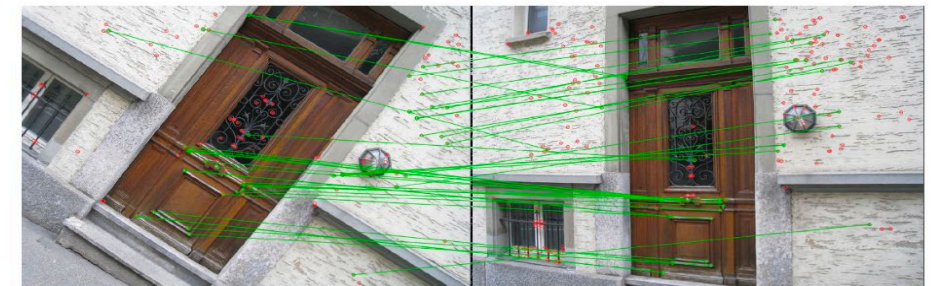
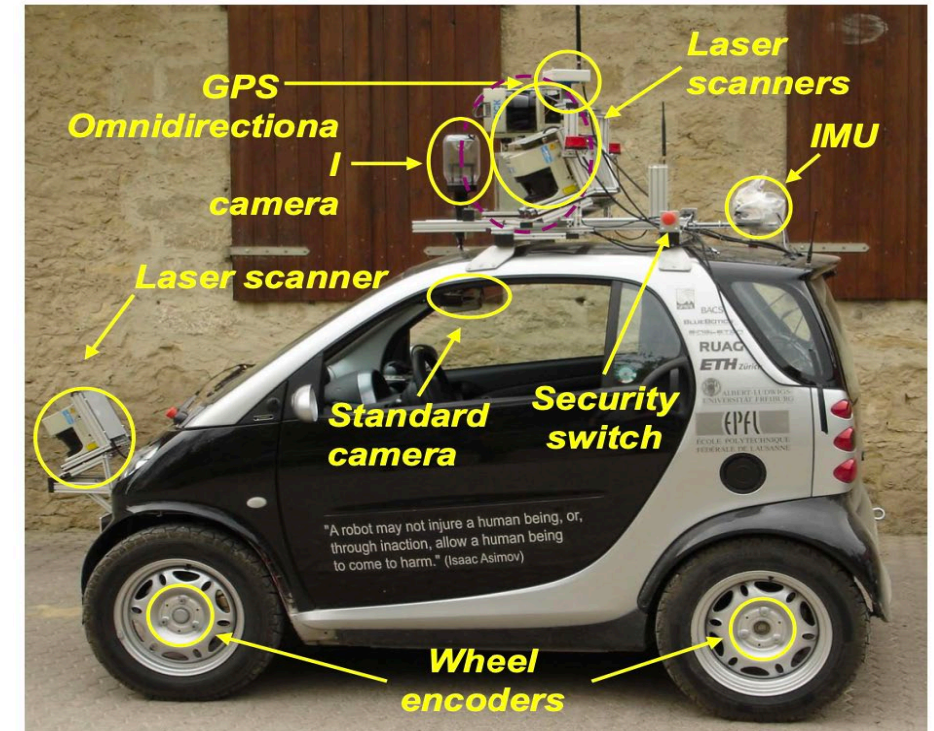
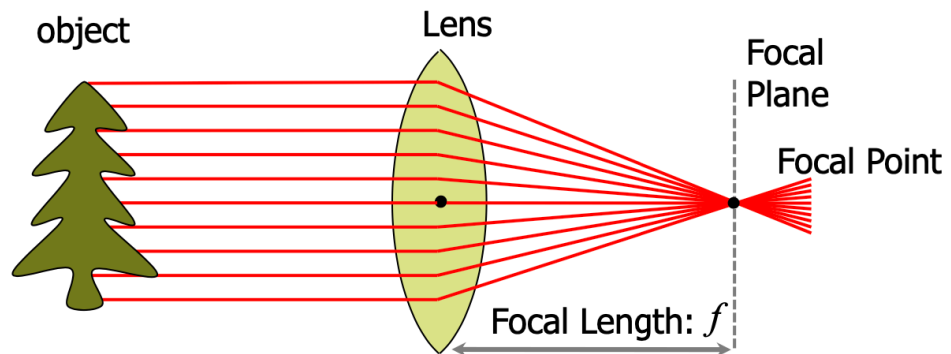
Perception: sensing

- Laser scanner

- time of flight

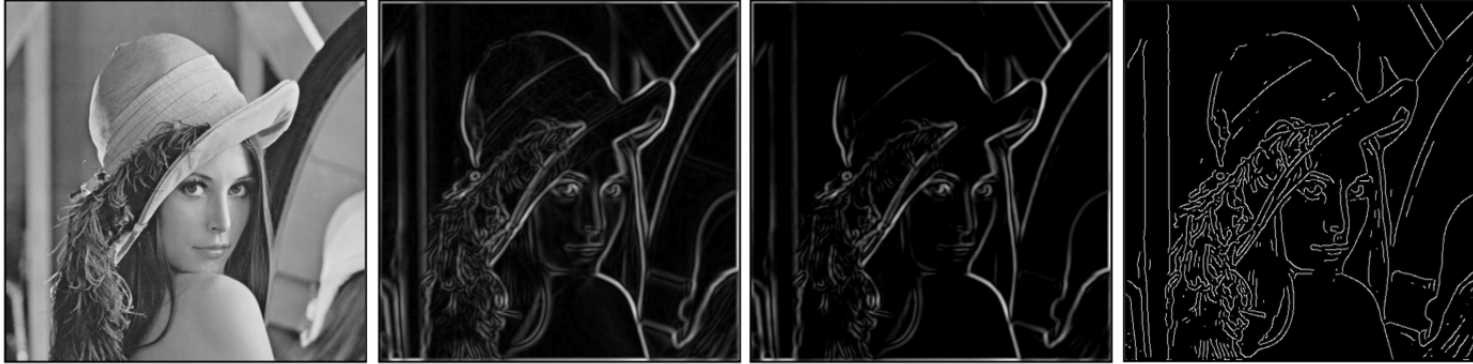


- Camera



<https://www.youtube.com/watch?v=yvgPrZNp4So>

Perception: information extraction



- Keypoint Features
 - Features that are reasonably invariant to rotation, scaling, viewpoint, illumination
 - FAST, SURF, SIFT, BRISK, ...



- Keypoint matching
 - BRISK

- Filtering / Edge Detection

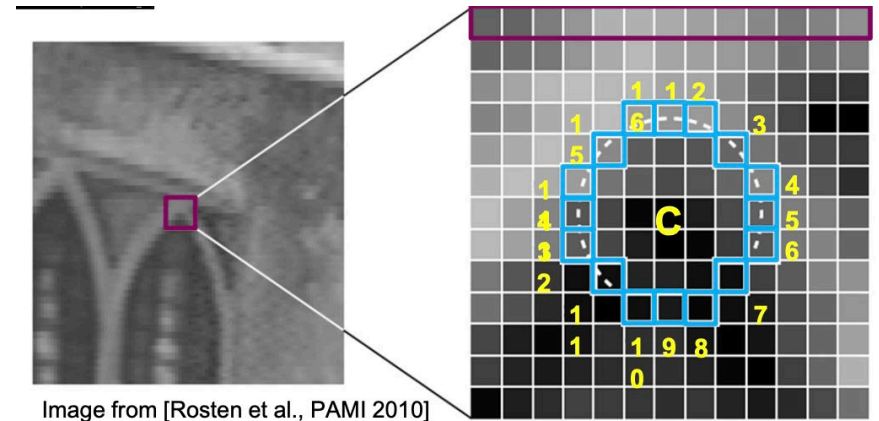
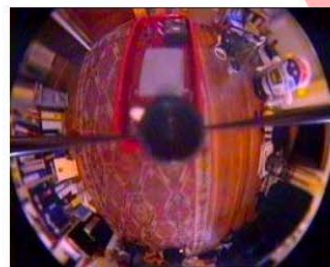
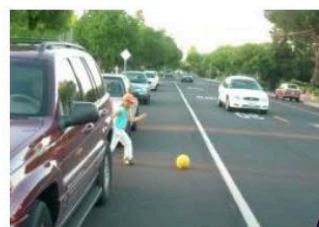


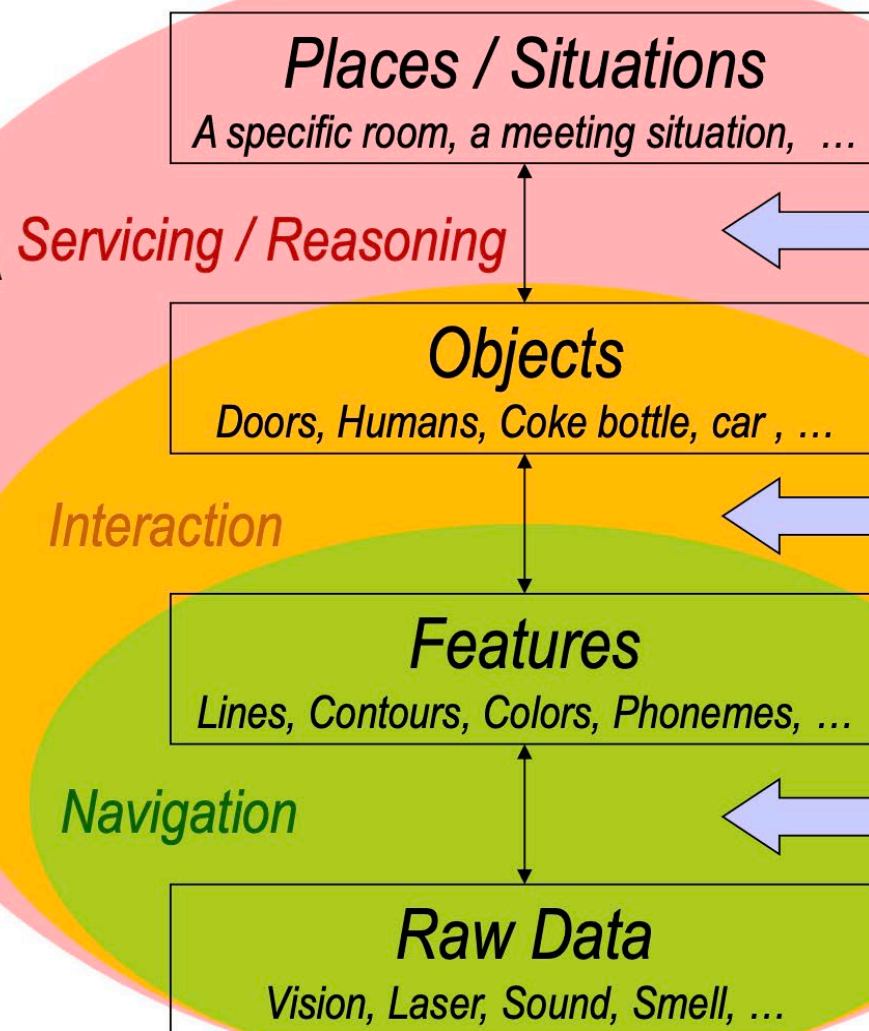
Image from [Rosten et al., PAMI 2010]

Perception

From Perception to Understanding



Fusing & Compressing Information



Servicing / Reasoning

Interaction

Navigation

• *Functional / Contextual Relationships of Objects*

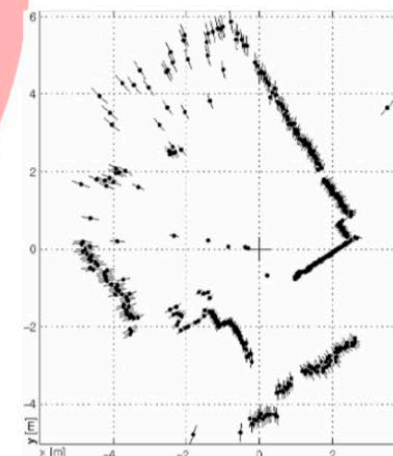
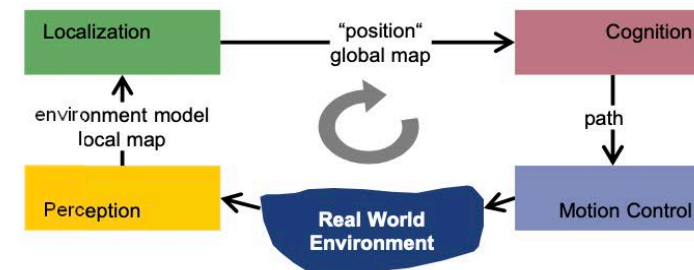
- imposed
- learned
- spatial / temporal/semantic

• *Models / Semantics*

- imposed
- learned

• *Models*

- imposed
- learned

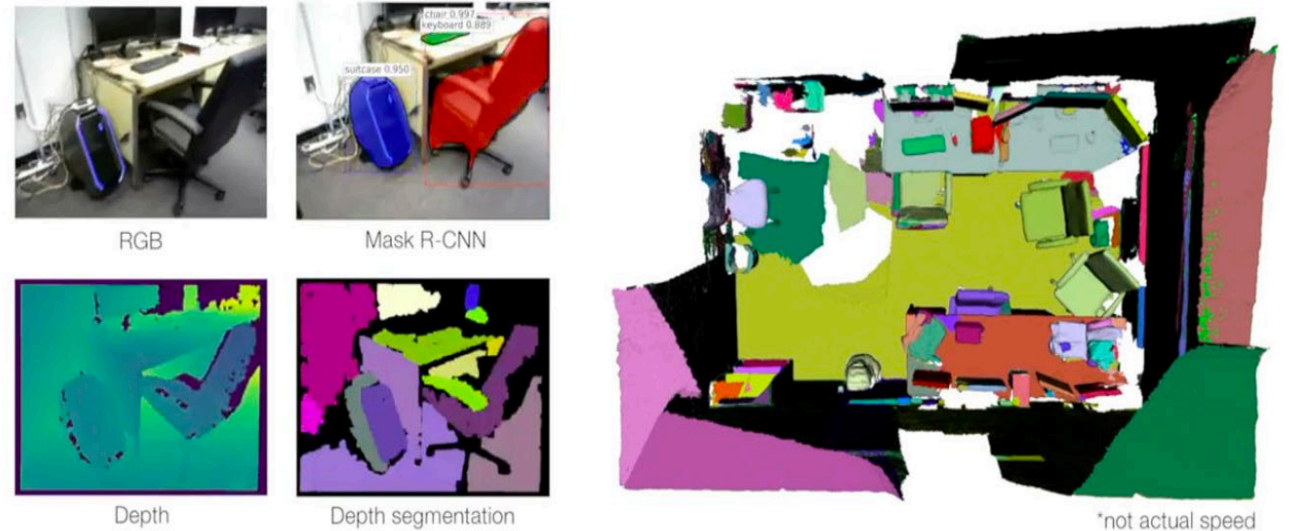


Perception

- Functional scene understanding via dense mapping at the level of objects
- A novel per-frame geometric-semantic segmentation scheme
- Detecting both recognized objects and previously unseen object-like elements

Open-source object-level mapping framework developed at ASL:

<https://github.com/ethz-asl/voxblox-plusplus>

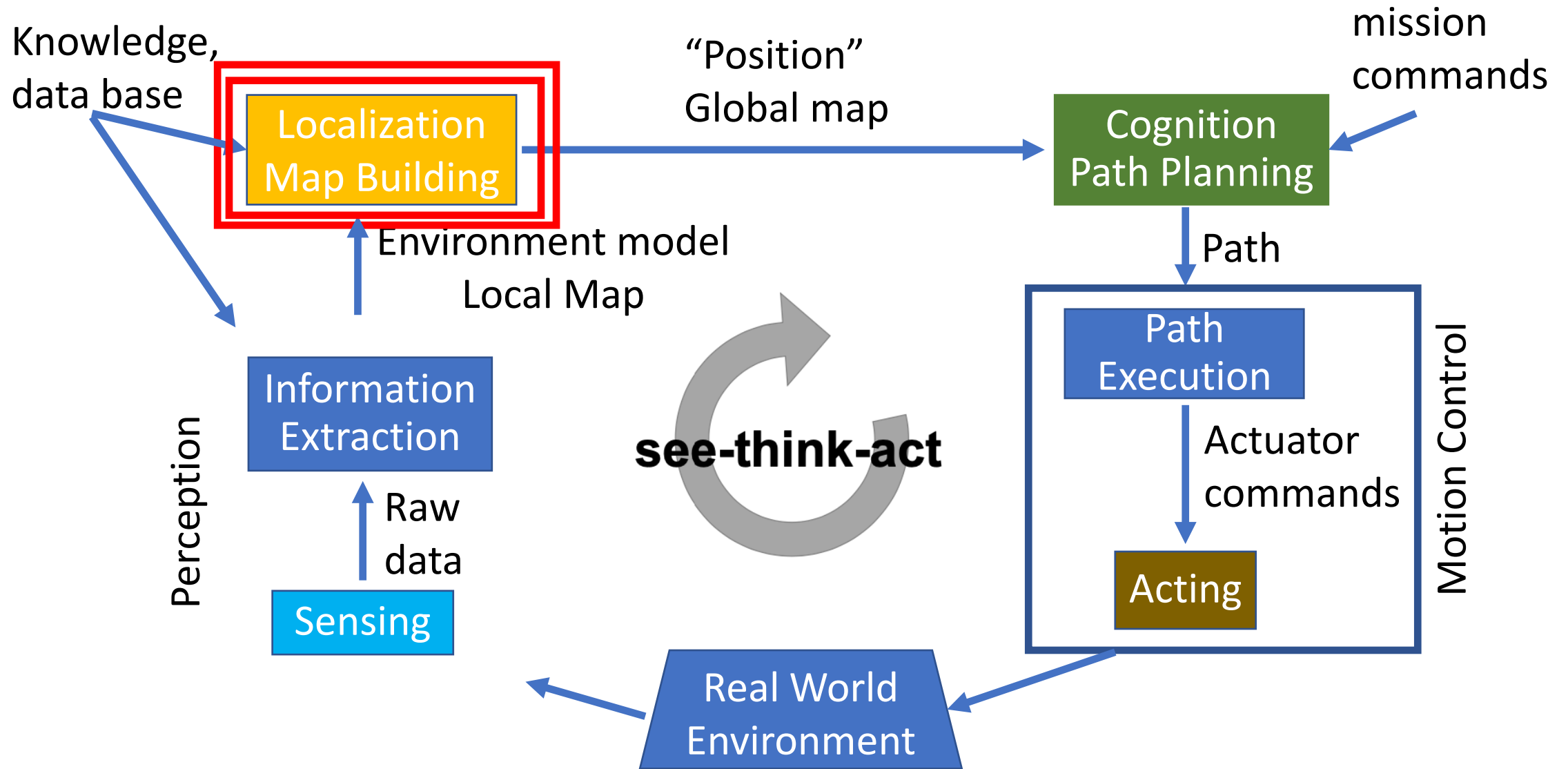


- Segments are incrementally integrated into a volumetric map, where a fusion strategy keeps track of the 3D geometry and location of the individual object instances detected.

M. Grinvald, F. Furrer, T. Novkovic, J. J. Chung, C. Cadena, R. Siegwart and J. Nieto. "Volumetric instance-aware semantic mapping and 3D object discovery." *RA-L* 2019.

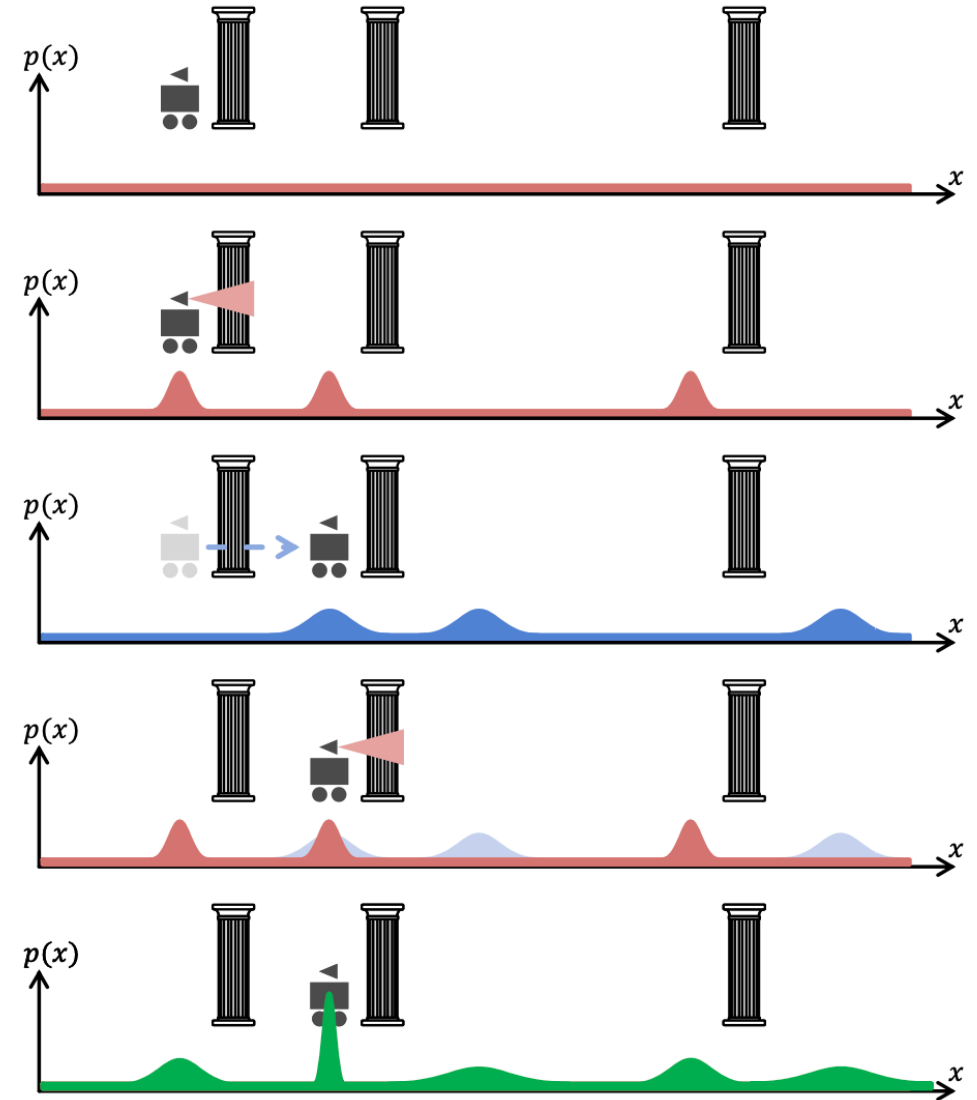
<https://youtu.be/JvI42VJmYxg>

The see-think-act cycle

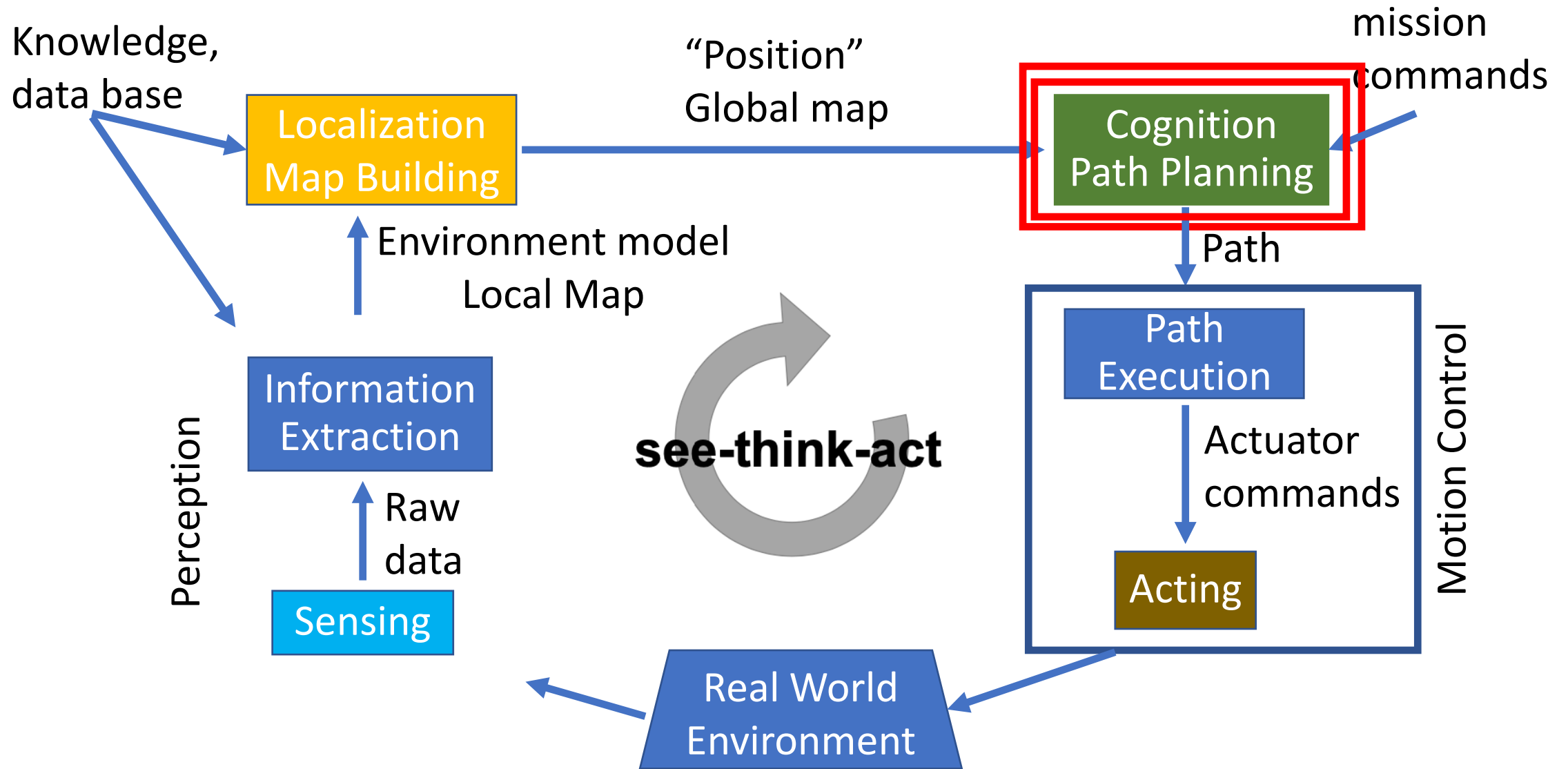


Localization: where am I?

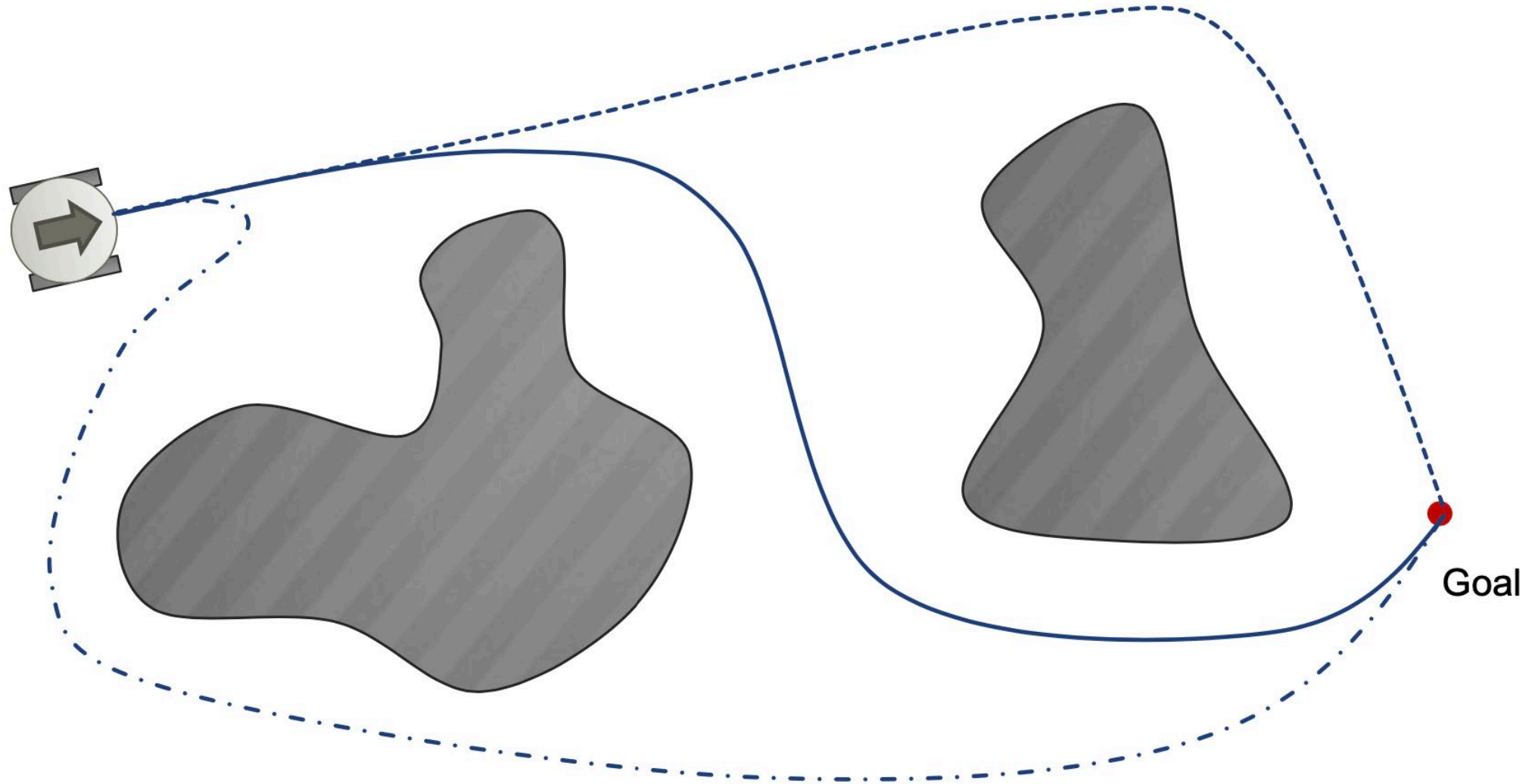
- SEE: The robot queries its sensors
→ finds itself next to a pillar
- ACT: Robot moves one meter forward
 - motion estimated by wheel encoders
 - accumulation of uncertainty
- SEE: The robot queries its sensors
again → finds itself next to a pillar
- Belief update (information fusion)



The see-think-act cycle

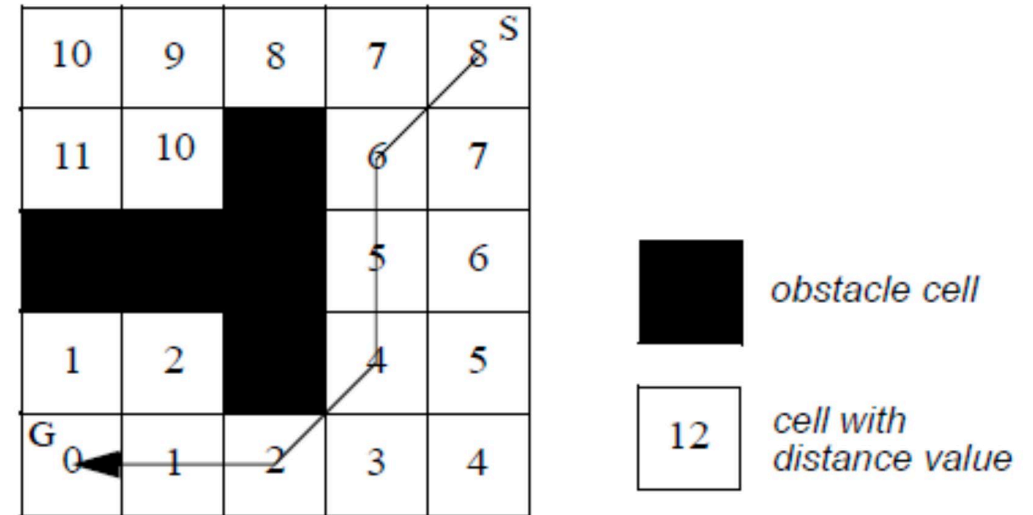


Cognition: Where am I going? How do I get there?

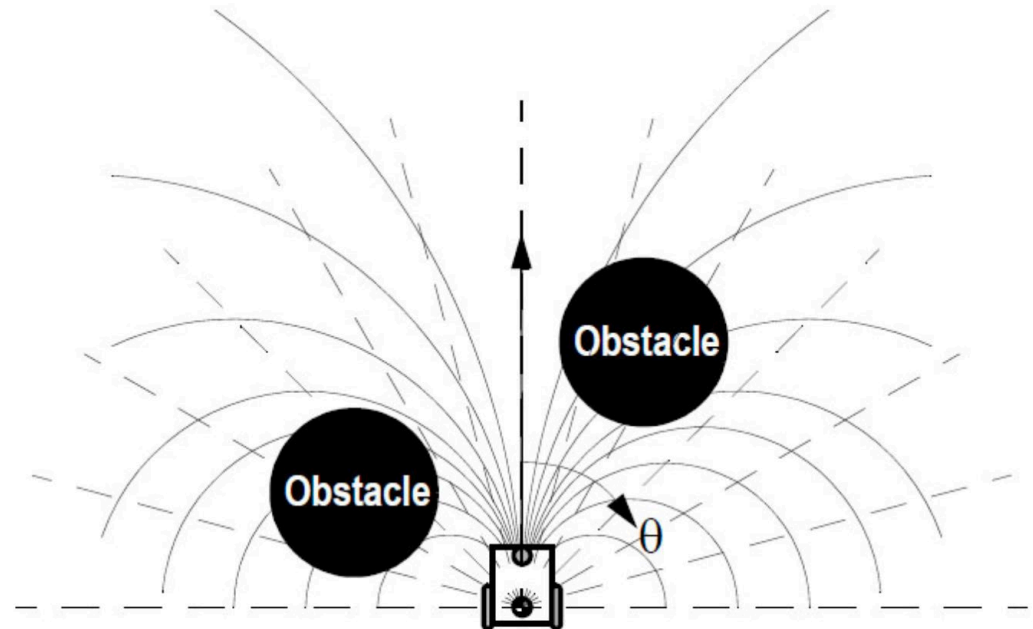


Cognition: Where am I going? How do I get there?

- Global path planning
 - Graph search

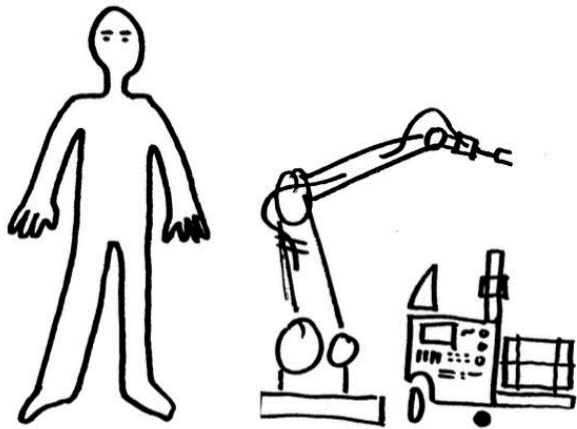


- Local path planning
 - Local collision avoidance

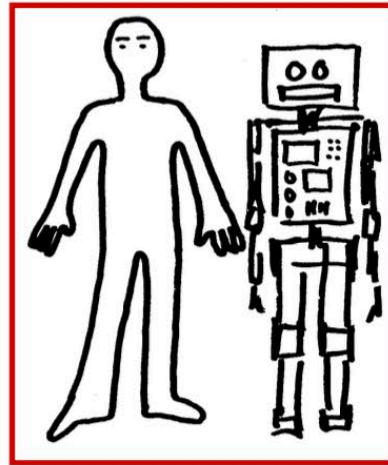


Next generation of Robots

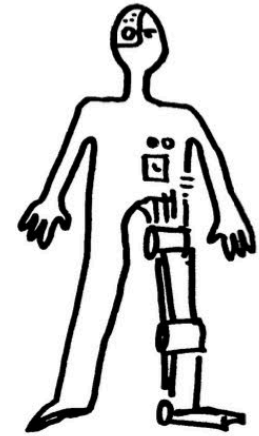
- Mobile, smart, connected, adaptive and closer to humans



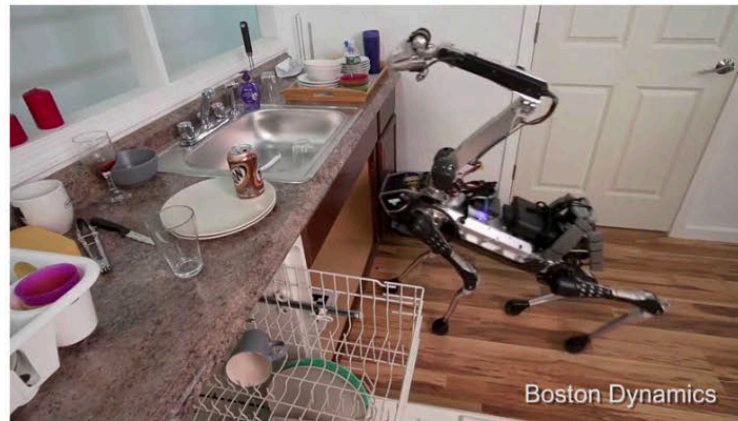
Industrial Robots



Service Robots



Cyborgs



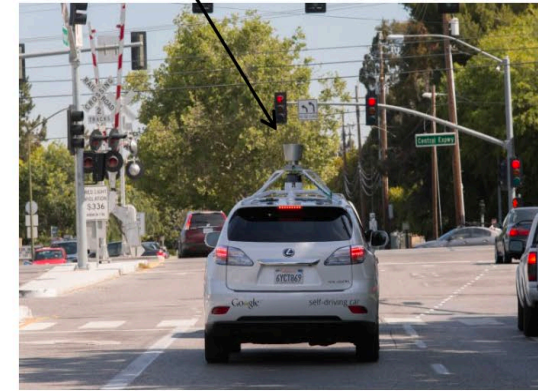
Boston Dynamics



Service Robots: Key Challenges

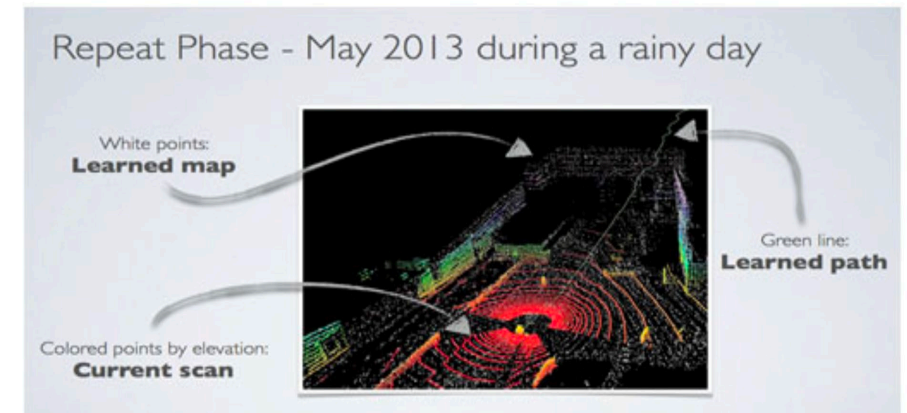
- Our real “physical” world is **multimodal**, very **diverse** and **complex**, we need robots that ...
- ... can **dealing** with **uncertain** and **partially available** information
- ... **see**, **feel** and **understand** their environment
- ... have **torque** and **force** control for tactile interaction (“soft robots”)
- ... offer **intuitive human-machine interfaces**
- ... **learn** and **adapt** every day
- → All this required AI/ML but even more, novel sensing, actuation and robot concepts!!

Today's Robots: 3D laser sensors



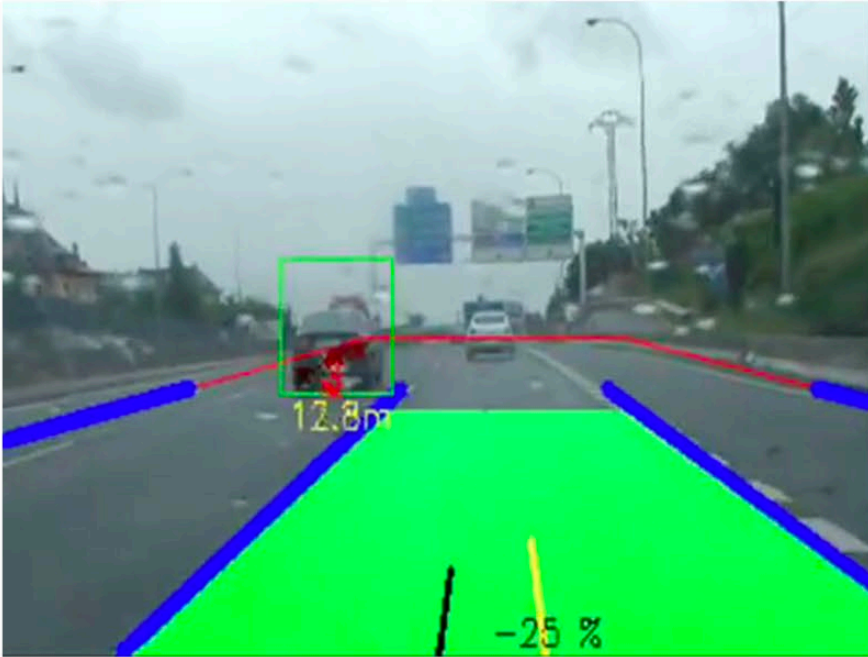
Expensive, complex and cumbersome

- Google Self-Driving Car Project (status summer 2015)
 - > 20 vehicles in use
 - > 2.7 million km, 1.5 million km in autonomous mode
 - > 11 accidents
 - No people insured
 - Non of them caused by car control algorithm



<https://www.youtube.com/watch?v=eJCR2TaeSFc>

Today's Robots: camera (lane tracking, ...)



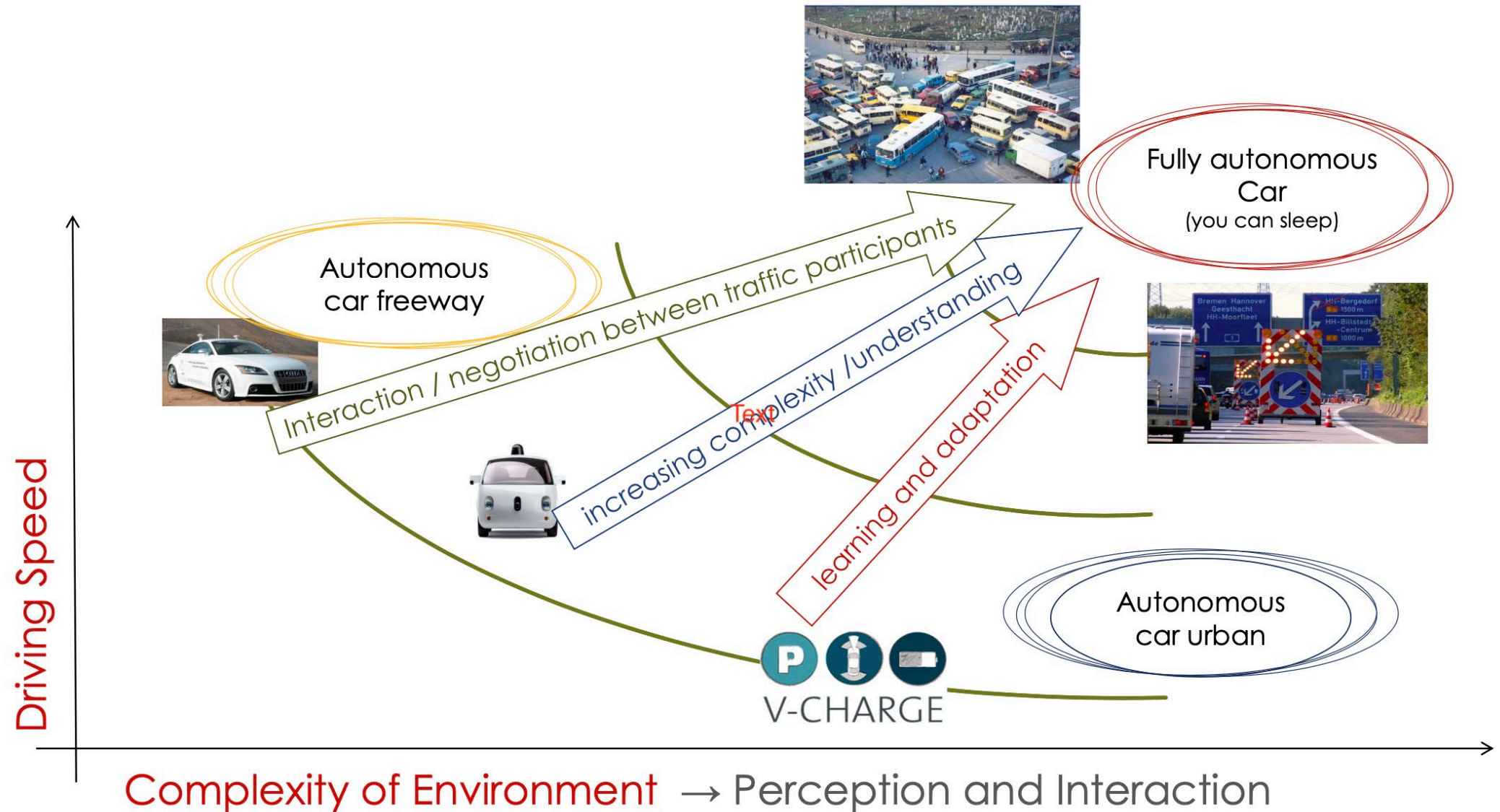
<https://www.youtube.com/watch?v=JmxDIuCllcg>



<https://www.youtube.com/watch?v=aGW4nRzx8lw>

- Detection and tracking of
 - Lanes
 - Street signs
 - Other cars

Today's Robots: Autonomous Cars Roadmap



Today's Robots: Vision only UAV navigation

- Swarm of small helicopters
 - Vision only navigation (one camera, GPS denied)
 - Fully autonomous with on-board computing
 - Feature based visual SLAM



<https://www.youtube.com/watch?v=0azJY7WY6tU>

<https://docs.px4.io/main/en/>

Navigation & Planning in Cluttered Environments



ZHEJIANG UNIVERSITY
COLLEGE OF CONTROL SCIENCE AND ENGINEERING

HUZHOU INSTITUTE OF
ZHEJIANG UNIVERSITY

| Swarm of Micro Flying Robots in the Wild

Xin Zhou, Xiangyong Wen, Zhepei Wang, Yuman Gao, Haojia Li, Qianhao Wang, Tiankai Yang, Haojian Lu, Yanjun Cao, Chao Xu*, Fei Gao*

Movie S1:

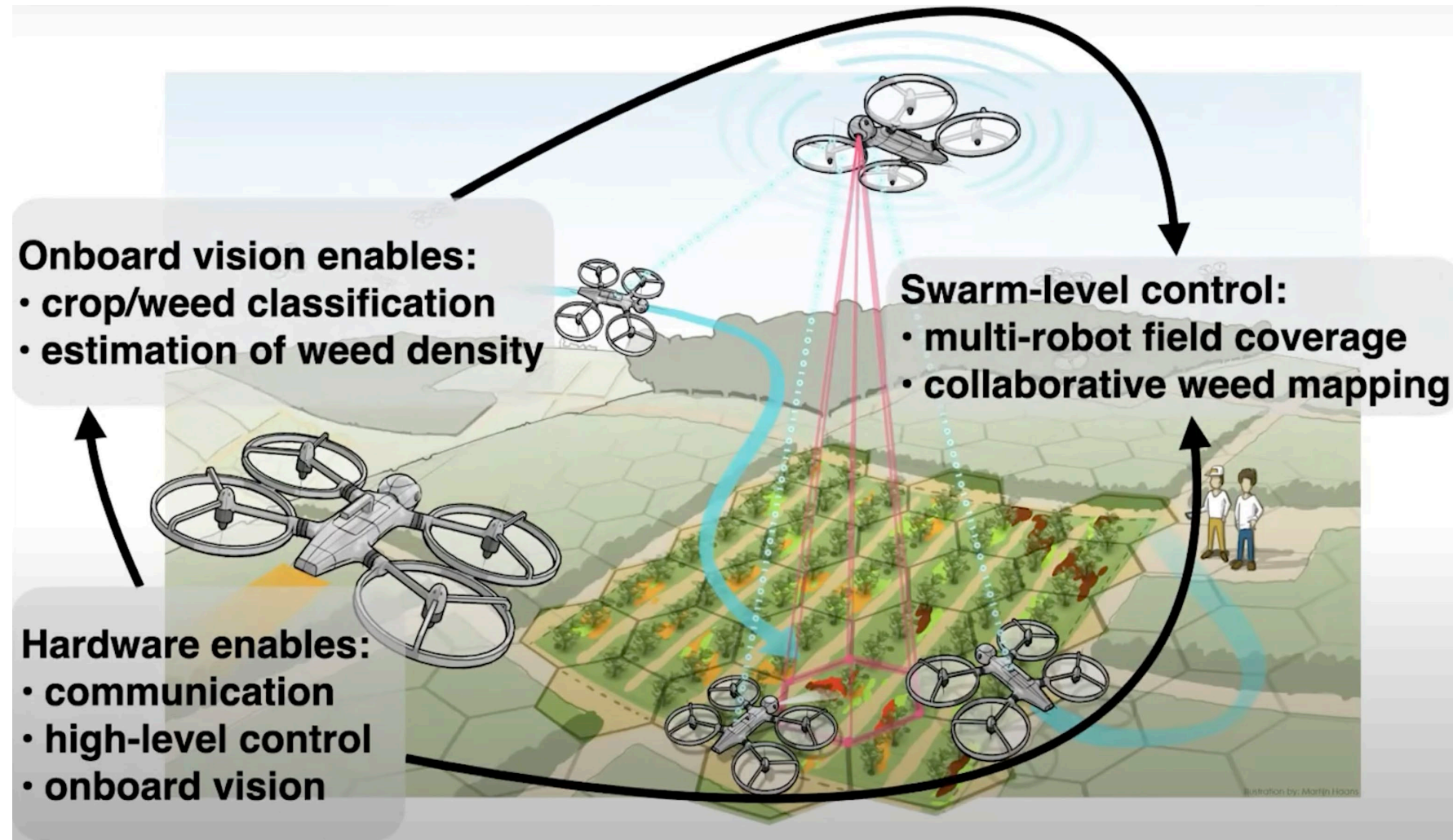
Swarm of Micro Flying Robots
in the Wild

FASTLAB
Field Autonomous System & compuTing



<https://www.youtube.com/watch?v=L0fJ0EHHfOA>

Flourish – Collaboration and Self-Organization of UAV Swarms for Precision Agriculture



Flourish – Why swarms?

- Parallelize operations
 - Higher efficiency
- Collaborative action
 - Higher accuracy
- Redundant systems
 - Higher robustness
- Decentralized algorithms
 - Higher scalability



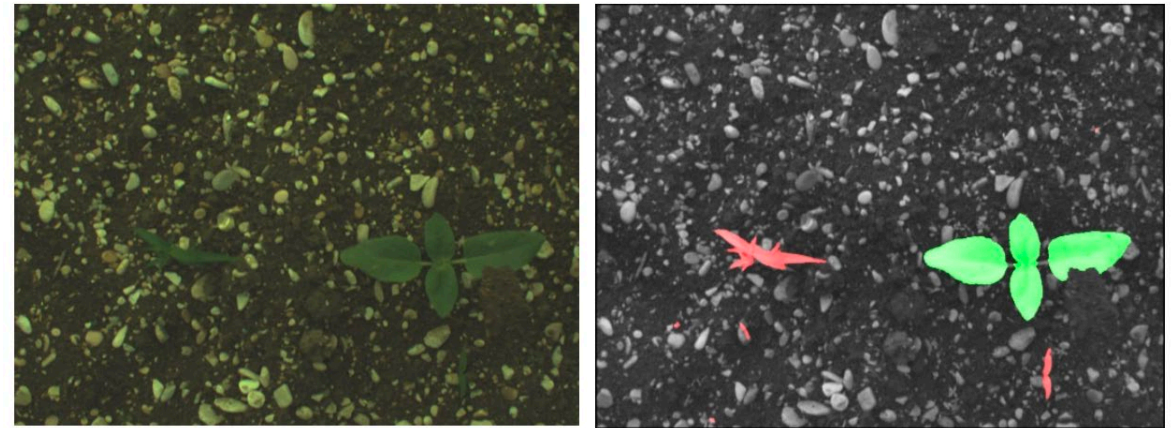
<https://www.youtube.com/watch?v=Ayz1zRE7RaI>

Flourish – Swarms for Precision Agriculture

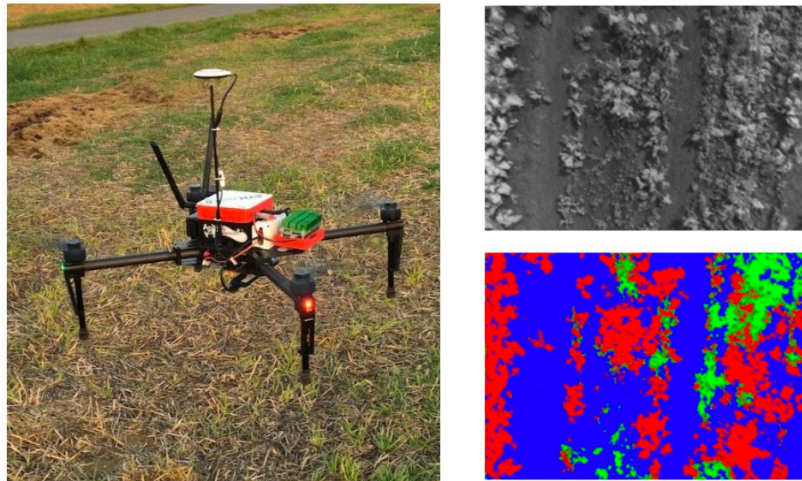
Autonomous UAV landing



Weed classification on UGV (Sunflower ~95% acc.)



UAV onboard weed detection



Automated spraying and stamping

